

Noise and meaning

Jens P. Blauert

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ence of nearby frequency components and in some cases for resolution of these components. Algorithms will be presented for obtaining a narrowing of either the usual positive peaks of the autocorrelation function or those of an inverted autocorrelation function [J. C. Brown and M. S. Puckette, Proc. 1987 Int. Conf. Comput. Music, Urbana, IL, 84-88 (1987)]. This latter function has some advantages for use in pitch track-

ing. These calculations will be applied to a variety of musical examples including the violin, flute, and piano, for the purpose of obtaining the pitch of digitized waveforms. These pitch tracking results will be presented for sampling rates of 10 K and 32 K and relative advantages will be discussed. [Work supported by a Brachman Hoffman Fellowship from Wellesley College.]

THURSDAY MORNING, 24 MAY 1990

204 KELLER, 8:30 A.M. TO 12:00 NOON

Session PP. Noise VI, Architectural Acoustics III, and Psychological Acoustics VI: Noise Quality I

David Lubman, Chairman

Hughes Aircraft, M.S. H425, Ground Systems Group, Building 618, P. O. Box 3310, Fullerton, California 92631

Chairman's Introduction—8:30

Invited Papers

8:35

PP1. History of noise quality in the automotive industry. Earl Geddes (Ford Motor Co., DPTC, 17000 Rotunda Dr., Dearborn, MI 48120)

This paper will present a brief look at the emerging philosophy of sound quality as viewed from within the automotive engineering community. A few examples of the concepts of sound quality will be discussed focusing on their relevance to noise control in vehicles. The paper will be primarily historical in content relating how the ideas of sound quality came to be recognized as an important factor in noise control for automobiles.

9:00

PP2. Psychoacoustics in context: Studying the listener and the noise. Rebecca N. Fleischman (Operating Sciences Dept., General Motors Res. Lab., Warren, MI 48090)

Psychoacoustic studies of product sounds can be improved by a better understanding of what the listener brings to the evaluation of noise. Noise is interpreted in the context of a system of knowledge and beliefs about products and their operation. When correlating acoustic noise factors with listener judgments, ask the following questions about the meaning and context of noise: What are the expectations that listeners have about noise? Listeners may expect certain product-specific sounds. They may judge product quality from specific noise features based on their beliefs about the relationship between the engineering or manufacturing of machines and their acoustic properties. What is the information content of a noise? Some components of a noise may be used as a diagnostic or to cue operation. What is the listening situation? Whether the listener is an active product operator or a passive bystander could mediate evaluations of noise quality. Examples of how these and other listener factors can influence noise judgments are presented, and strategies for incorporating them in psychoacoustic studies are discussed.

9:25

PP3. Noise and meaning. Jens P. Blauert (Lehrstuhl fuer Allgemeine Elektrotechnik & Akustik, Ruhr-Univ. Bochum, D-4630 Bochum, West Germany)

Noise is undesired sound. The decision, however, of whether a sound is desired or undesired, is taken by the listener in the specific situation of sound exposure. A sound that is noise for one listener, may be a useful, information carrying signal for another listener and, moreover, each listener may change his mind at any time depending on the circumstances. Consequently, once a noise level is reduced so far that a physiological danger for the listeners' ear is no longer present, a further reduction of the level may not be the adequate measure to

further decrease the negative impacts of the noise. Instead, a careful redesign of the complete acoustical environment at the listeners' position may be required. To this end, proper, ear-adequate methods for noise measurement and noise quality evaluation are needed. In this talk, some developments along these lines, as currently being discussed in Germany, will be reviewed and evaluated.

9:50

PP4. Ear-adequate spectral analysis for noise evaluation. Klaus Genuit (HEAD Acoustics GmbH, D-5120 Herzogenrath 3, West Germany)

In numerous sound situations, sound quality cannot be determined by means of conventional measuring techniques. The frequent discrepancy between subjective judgment of sound events yielded by standard measurement procedures, can be explained by the difference between a sound situation as received by a single omnidirectional microphone, on the one hand, and as received by the human hearing, on the other hand. A new procedure using a dummy head and a special binaural analyzer has been developed for spectral analysis that realizes frequency and time resolution in the entire audio spectral range. This is comparable to the properties of human hearing, which is able to perceive quickly changing time structures and minor frequency differences at the same time. New approaches for ear-adequate noise evaluation in order to judge noise quality can be achieved by means of psychoacoustic properties of human hearing and binaural signal processing. The results of these investigations show that human hearing, which has two auditory paths with a complex directional characteristic and a binaural pattern recognition, gives a different judgment of noise quality as compared to conventional measurement methods, especially in complex sound situations with a spatial distribution of several sound sources and a sound-pressure level below 85 dB.

10:15

PP5. A method for evaluating interior sound quality of automobiles. Tsuyoshi Yamashita, Yasuo Ishii, Mitsuo Nakamura (Honda R&D Tochigi Center, Shimotakanazawa, Haga, Tochigi, 321-33 Japan), and Otoichi Kitamura (Osaka Univ. of Arts, Osaka, 585 Japan)

In the past, the A-weighted sound-pressure level has been used to evaluate the noise level inside a vehicle cabin, since great importance is attached to reducing loudness. This report is an attempt to analyze sound "quality," which is difficult to express solely in terms of loudness, using multivariate analysis performed on the results of a 20-items semantic differential method survey. The results confirm that a vehicle's image bias has a greater effect on responses than actual differences in vehicle noise, thus clearly demonstrating the importance of a blindfold test. Proceeding from this result, blindfold tests were performed using the sound recorded from actual vehicles, both ordinary passenger cars and sports cars, and principal component analysis was performed on the results. Three principal tone factors were extracted: the "beauty factor," the "powerfulness factor," and the "metallicness factor." The principal tone factors for automobile interior noise are similar to the results of studies done in other fields.

10:40

PP6. Generating vehicle sounds synthetically. Anthony J. Champagne, James C. Rogers (Electr. Eng. Dept., Michigan Technol. Univ., Houghton, MI 49931), and Ray Wales (Ford Motor Co., Allen Park, MI 48121)

Synthetic vehicle sounds can be used for subjective listening tests. Computer models provide a great deal of flexibility while mechanical systems are not easily modified for such tests. Autoregressive (AR) parametric models were developed using real data for training. With this approach relatively few parameters are needed to model the vehicle sounds. Furthermore, arbitrarily long data sequences can be produced from these models for subjective evaluations by listeners. Interior recordings in two 4-cylinder engine cars and in two 6-cylinder engine cars were used as training data. It was found that signals with more complex spectra were modeled most successfully from the point of view of a listener. Signals with simple spectra (one dominant frequency component) were more difficult to model. Modeling difficulty manifests itself in amplitude variability with time in the sounds produced with the synthetic data. Measures to counter these effects have been met with some success. [Work supported by Ford Motor Company.]

11:05

PP7. Signal processing techniques for identifying sound quality parameters. P. Laux and P. Davies (Ray W. Herrick Labs., School of Mech. Eng., Purdue Univ., West Lafayette, IN 47907)

Importance is now being placed on the *type* of sound that machinery makes rather than simply on the noise level it produces. Two reasons for this emphasis are: the adverse affect an unpleasant acoustic environment can